

Productivity of exceptional and unnatural phonology: Evidence from Catalan consonant alternations

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Abstract

This study addresses two important research areas in theoretical phonology: exceptional and “unnatural” phonology. We carried out two wug-test experiments with native speakers of Central Catalan in which they provided inflected forms for nonce words, in paradigms attesting four phonological processes: /n/-deletion, /r/-deletion, /nt/-cluster simplification, and /z/ ~ [tʃ] alternation. Our experimental findings support the following claims: (i) Phonological processes can be productive even when they have numerous lexical exceptions. (ii) Speakers confronted with exceptional phonology tend to frequency-match the patterns of the lexicon. (iii) For two forms of “unnatural” phonology, opacity and saltation, speakers also productively extend the lexical pattern, but they underapply these processes relative to their lexical frequencies.

Keywords: Catalan phonology, wug test, productivity, frequency-matching, opacity, saltation

Word count: 13162

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1. Introduction

The sound pattern of Catalan² has for decades played an important role in the development of phonological theory, notably in the work of Mascaró (1975), Wheeler (1979, 2005), and Bonet & Lloret (1998). In this paper we report on a “wug test” (Berko 1958) targeting a small set of consonant alternations that arise at the right edges of Catalan stems. Our goal is to use these alternations to probe phonological patterns that are of theoretical interest, particularly those that have been argued to pose learnability difficulties. We will examine two domains: **exceptionful phonology**, and two forms of “**unnatural**” **phonology**, i.e. opaque process interaction and saltation. The specific research questions, along with the phonological patterns we will use to address them, are given in (1).

(1) *Research questions*

- a. **Productivity of exceptionful phonology:** When a phonological process of a language has lexical exceptions, and speakers are tested in ways that invite them to extend the process productively, do they extend the process to novel forms, or do they suppress it? Moreover, if speakers do extend a process, do they also *frequency-match* (Zuraw 2000; Ernestus & Baayen 2003) the pattern? By this we mean that their answers in the aggregate track the different application rates seen in different phonological contexts. Below, we consider productivity and frequency-matching in two well-studied phonological processes of Catalan, final /n/-deletion and final /ɾ/-deletion.
- b. **Productivity of unnatural phonology:**
 - i. *Opacity:* Can speakers productively extend a phonological pattern that is *opaque*, in the sense of Kiparsky (1973)? It has been suggested that phonological opacity leads patterns to be less productive (Mayer 2025) or entirely unproductive (Sanders 2003). Here, we study productivity for the counterfeeding of /n/-deletion by /nt/-cluster simplification in Catalan, which is exceptionless in the lexicon.
 - ii. *Saltatory alternations:* In “saltation”, segment A alternates with B, passing over stable intermediate C (Łubowicz 2002, Ito & Mester 2003). It has been suggested that saltation, like opacity, is limited in its productivity (Hayes & White 2015). We focus on a saltatory alternation of Catalan whereby /z/ is realised as [tʃ], leaping over intervening /f/. This is also an exceptionless process.

These alternations are described in more detail below. In the paradigms given, the addition of the feminine [-ə] suffix prevents the stem from being word-final. It is only in the masculine, where

² Catalan is a Western Romance language; speakers of Catalan may be found in Catalonia, Valencia, the Balearic Islands, Andorra, parts of southern France, and the city of Alghero on Sardinia. Our focus here is on Central Catalan, spoken in Barcelona and neighbouring regions.

the right-edge of the stem is “exposed” to the phonology, that the various processes apply. The examples below are drawn from the presentations in Mascaró (1975) and Wheeler (2005).

(2) *Alternations chosen for study*

a. */n/-deletion*

/san/ → ['sa] ‘healthy.M.SG’; cf. F.SG ['san-ə]

b. */r/-deletion*

/klar/ → ['kla] ‘clear.M.SG’; cf. F.SG ['klar-ə]

c. */nt/-cluster simplification*

/sant/ → ['san] ‘holy.M.SG’; cf. F.SG ['sant-ə]

d. */ʒ/ → [tʃ] alternation*

/bɔʒ/ → ['bɔtʃ] ‘crazy.M.SG’; cf. F.SG ['bɔʒə]

Catalan stands out for the care with which the data have been addressed in traditional generative analysis; and as such it has been widely employed in textbooks (Kenstowicz & Kisseberth 1979; Odden 2005) and computational study (Cotterell *et al.* 2015; Shilen & Wilson 2022; Rasin *et al.* 2021; Wang & Hayes 2025). However, purely analytic research cannot tell us much about the productivity of the phonological processes involved. Although some insight can be obtained from loanword adaptation, notably in work by Pons-Moll (2015), experimentation is the most comprehensive and systematic way to evaluate the productivity of phonological patterns and contribute to long-standing theoretical debates in a meaningful way.

1.1 Outline

In §2, we describe in more detail the consonant alternations of Catalan on which we focus. Our description relies on a lexical corpus of Catalan paradigms which permits us to assess the numerical representation of rival patterns. §3 covers the design of our experiments: participants, stimuli, and experimental procedure. §4 gives the results, assessing how they bear on the research questions of (2), and offering our speculations concerning the less-expected outcomes. §5 offers a summary and directions for further research.

2. The phonology of stem-final consonants in Catalan

We rely heavily on the descriptions and analyses given in Mascaró (1975) and Wheeler (2005). Wheeler (2005) covers several Catalan dialects, but here we will focus entirely on Central Catalan, the variety spoken in Barcelona and surrounding regions.

In this section we employ, simply for terseness, a rule-based analysis (Chomsky & Halle 1968).

2.1 Database

In order to address the question of frequency-matching, we need quantitative assessment of how often the various patterns are represented in the lexicon, hence a lexical survey. The survey is intended to approximate the frequency patterns that a Catalan speaker would likely have encountered while learning the language, and hence lets us assess in a rough way whether speakers are engaged in frequency-matching. As we will see below, it is also important to know how many words attest the phonological process under discussion, since this influences speakers' behaviour as well.

We focused on one particular paradigm in which the alternations we study are particularly abundant, namely the inflection of noun and adjective stems for gender and number. In Wiktionary (www.wiktionary.org; 185K words), such stems are generally reported with their full gender/number paradigm; for instance, the paradigm for 'small' is as given in (3).

(3) *Sample paradigm for regular Catalan nouns and adjectives*

a. Masculine singular	∅	[pə'tit]
b. Masculine plural	[-s]	[pə'tit-s]
c. Feminine singular	[-ə]	[pə'tit-ə]
d. Feminine plural	[-ə-s]	[pə'tit-ə-s]

This is the regular pattern; there are also irregular paradigms, which turn out to play an important role here. In particular, in some cases, the masculine takes an [-u] ending (e.g., [mu'rɛn-u] 'brunet.M.SG'), or a [-ə] ending (e.g., [pəriu'dist-ə] 'journalist.M.SG').

We used *Wikiextract* (Ylonen 2022) to obtain all such four-member paradigms from Wiktionary, then removed all words not known to the native-speaker second author ($n = 619$), since such forms are less likely to be known to our participants. With these deductions our list came to 5761 paradigms. Several of the alternations we study are not expressed in Catalan orthography, so we needed IPA transcriptions. These were taken from Wiktionary where available; else they were generated with the *catalan2ipa* package (Groß 2019). All transcriptions were checked by the native-speaker author.

A few forms show optionality in the relevant phonology; for example our source gives two possibilities for the masculine form of [iŋ.ku'lo.rə] ‘colorless.F.SG’, namely [iŋ.ku'lo] and [iŋ.ku'lor] ‘colourless.M.SG’. In calculating application rates for such cases, we counted each variant as 0.5.

2.2 Exceptionful phonology: /n/ and /r/-deletion

Catalan displays two stem-final deletion processes - /n/ and /r/-deletion - which apply frequently but far from categorically. Deletion is the predominant pattern in the lexicon, but there are also numerous lexically-specific exceptions.

Both /n/ and /r/-deletion target word-final segments, though /r/-deletion applies in addition to segments before final /s/ (e.g., /klar + s/ → [klas] ‘clear.M.PL’ but /plan + s/ → [plans] ‘flat.M.PL’). In the nominal/adjectival paradigms studied here, /n/ and /r/-deletion result in alternations, as deletion takes place in the unsuffixed masculine forms but not in feminines. Examples of both processes are given in (4b).

(4) a. /n/ and /r/-deletion

$n \rightarrow \emptyset / V ___]_{\text{word}}$	‘Delete postvocalic [n] word-finally’
$r \rightarrow \emptyset / ___ (s)]_{\text{word}}$	‘Delete [r] word-finally and before final [s]’

b. Derivations

<i>‘flat.M.SG’</i>	<i>‘flat-F.SG’</i>	<i>‘clear.M.SG’</i>	<i>‘clear-F.SG’</i>	
/plan/	/plan-ə/ ³	/klar/	/klar-ə/	underlying representations
∅	—	∅	—	/n/-deletion
—	—	—	—	/r/-deletion
[ˈpla]	[ˈplanə]	[ˈkla]	[ˈklarə]	surface representations

As Mascaró (1975) and Wheeler (2005) point out, /n/ and /r/-deletion both involve a fair number of exceptions. For instance, we might expect the UR /nən/ ‘child.M.SG,’ (F.SG [ˈnən-ə]) to surface in the masculine as *[ˈnɛ], but in fact it is pronounced [ˈnən]. Likewise, the UR /pur/ ‘pure.M.SG,’ (F.SG [ˈpur-ə]) surfaces as [ˈpur] and not *[ˈpu]. Pairs like /plan/ vs. /nən/ and /klar/ vs. /pur/ demonstrate that neither /n/ nor /r/-deletion are entirely predictable.

The presence of exceptions naturally raises the question of whether these alternations are productive in Catalan. Indeed, Wheeler (2005) and Pons-Moll (2015) suggest that they are not,

³ We adopt /-ə/ as the UR for the [-ə] suffix simply for convenience. Since /e/, /ɛ/ and /a/ all reduce to schwa in stressless position, any of them could serve as the UR. For discussion, see Wheeler (2005:§2.3).

observing in particular that they usually fail to apply in loanwords (e.g., *xaman* ‘shaman’ [ʃəˈman], *amateur* ‘amateur’ [ə.məˈteɪ]). The issue bears on the general question of the conditions under which phonological patterns that suffer from a substantial number of exceptions can be productive (see, e.g. Hooper 1976; Kiparsky 1982; Zuraw 2000; Bybee 2001; Pierrehumbert 2006; Yang 2016). It will be seen that our experimental results support at least some degree of productivity for /n/ and /r/-deletion.

As with exceptionality elsewhere (Zuraw 2000), exceptionality for /n/ and /r/-deletion is *patterned*, in the sense that one can locate particular environments in which application is especially frequent or especially unusual among the words of the existing lexicon. Because of this patterning, the distribution of /n/ and /r/-deletion is not random. Importantly for this study, experimental work shows that in other languages, participants are able to use such patterning to guide their responses in a wug-test—in other words, they show a tendency to *frequency-match* the lexicon (Hayes *et al.* 2009:826; Becker & Gouskova 2016; O’Hara 2020; Song & White 2022; Gouskova 2025). Thus, in assessing the productivity of the Catalan alternations, we want to bear in mind the pattern of structured exceptionality that we lay out immediately below.

We consider four environments that affect the rate of /n/ and /r/-deletion in the lexicon, shown in (5). Most of these were noticed earlier by Mascaró (1975) or Wheeler (2005).

(5) *Environments for /n/ and /r/-deletion*

- a. **Particular suffixes.** In words formed with /-in/, a frequent adjectival suffix, /n/-deletion applies 100% of the time (105/105 cases in our corpus). Likewise, /r/-deletion applies with perfect regularity (205/205, 100%) in forms with the suffix /-dor/ ‘agentive’. Some examples are [məʎurˈk-in-ə] ~ [məʎurˈk-i] ‘Mallorcan-F.SG/M.SG’ and [ədministrə-ˈdor-ə] ~ [ədministrə-ˈdo] ‘administrator-F.SG/M.SG’.
- b. **Penultimately-stressed stems.** Various stems of Catalan have stress on their second to last syllable, hence penultimate word stress in the masculine and antepenultimate in the feminine; e.g. [əwˈtəktun-ə] ~ [əwˈtəktun] ‘autochthonous-F.SG/M.SG’ and [ˈprɔspər-ə] ~ [ˈprɔspər] ‘prosperous-F.SG/M.SG’ both have penultimately-stressed stems. In such forms, deletion is unusual (1/27 in our corpus for /n/, or 3.7%; 2/24 for /r/, 8.3%). The examples just given are, thus, typical.
- c. **Monosyllabic stems.** These often permit deletion, but in only about half (8/15 for /n/, 53.1%; 3/7 for /r/, 42.9%) of the cases. Examples of deleting and non-deleting monosyllables were noted above: [ˈplan-ə] ~ [ˈpla] ‘flat-F.SG/M.SG’, but [ˈnɛn-ə] ~ [ˈnɛn] ‘child-F.SG/M.SG’; and [ˈklar-ə] ~ [ˈkla] ‘clear-f./m.’ vs. [ˈpur-ə] ~ [ˈpur] ‘pure-F.SG/M.SG’.

- d. **Other.** The remaining category, a large one, is “elsewhere,” meaning: /n/ or /r/ in final position of a polysyllabic stem, with a stressed stem-final syllable, and neither in /-in/ nor /-dor/. Here, deletion is frequent (390/410 for /n/, 95.1%; 250/256 for /r/, 97.7%), but not the 100% observed with /-in/ and /-dor/. Examples are [kətə'lan-ə] ~ [kətə'la] ‘Catalan-F.SG/M.SG’ and [pri'mer-ə] ~ [pri'me] ‘first-F.SG/M.SG’.

As an example of what guided our experimentation on frequency-matching, we offer a specific prediction: if our test includes a set of feminine wug words that have stem-final stress and are unsuffixed (such as our wug words [prə'ton-ə] or ['brɔn-ə]), we expect the group of experimental participants collectively to give masculine responses involving deletion more frequently when the wug stem is polysyllabic (like [prə'to] for [prə'ton-ə]) than when it is monosyllabic (like ['brɔ] for ['brɔn-ə]), thus matching the difference in lexical frequencies. More generally, the sample of responses from the participants, taken as a whole, will be the result of their stochastically matching the frequency patterns given in (5).

There is a substantial body of work that attempts to explain the cases where speakers deviate from precise frequency-matching. For example, Becker *et al.* (2012) suggest that participants deviate from frequency-matching in order to respect a UG bias disfavouring alternations in initial syllables; for other cases see e.g., Zhang *et al.* (2011), Moreton & Pater (2012), and Hayes & White (2013). While establishing principles of learning bias is of great theoretical interest, our goal is more modest, namely just to establish whether exceptional processes such as /n/ and /r/-deletion can be productive and if there is a tendency toward frequency-matching in Catalan.

2.3 Unnatural phonology

There are also two stem-final processes in Catalan which, as discussed in the Introduction, violate proposed notions of phonological naturalness: the counterfeeding of /n/-deletion by /nt/-cluster simplification is a classic instance of opacity (Kiparsky 1973), and the alternation of stem-final /ɜ/ with [tʃ] (not *[ʃ]) is a case of saltation. Investigation of these patterns allows us to check whether speakers productively extend these possibly unnatural alterations. If they do not productively extend the alternations, or they only do so partially, we might expect speakers to produce the corresponding transparent forms (for opacity), or forms that avoid saltation by employing the “leaped over” segment in the actual output. As it turned out, our experimental data reveal substantial variation among alternatives, as well as other participant responses.

2.3.1 Opacity and /nt/-cluster simplification

In broad terms, word-final homorganic clusters in Catalan are simplified; the full pattern is given in detail by Mascaró (1975) and Wheeler (2005). Here, we focus on a single -cluster, /nt/. When no suffix follows, underlying /nt/ is rendered as [n]. An example is the stem for ‘holy’, /sant/, which surfaces intact in the feminine ['sant-ə] but with loss of /t/ in masculine ['san]. Our

database has 40 stems with final underlying /nt/, and every one of them surfaces with final [n].⁴ More generally, irrespective of paradigm structure, Catalan does not tolerate word-final [nt]. /nt/-cluster reduction is stated and exemplified in (8), which also gives the crucial ordering relationship with /n/-deletion.

(8) a. /nt/-cluster reduction

$t \rightarrow \emptyset / n ____]_{\text{word}}$			‘Delete [t] word-finally after /n/’
b. ‘holy-F.SG’	‘holy.M.SG’	‘good.M.SG’	
/sant-ə/	/sant/	/bɔn/	underlying representations
—	—	∅	/n/-deletion
—	∅	—	/nt/-cluster reduction
['santə]	['san]	['bɔ]	surface representations

As can be seen, in rule-based phonology, /nt/-cluster simplification *counterfeeds* (Kiparsky 1968) /n/-deletion, since [n] rendered word-final by /nt/-cluster reduction is never deleted (8).

Irrespective of which analytic approach we take, the key point is that this data pattern involves *opacity* (Kiparsky 1973), since /nt/-cluster reduction creates surface exceptions to /n/-deletion. Starting with Kiparsky’s original work, it has been suggested that opacity, at least in some of its forms, is impossible (Sanders 2003) or merely difficult (Bowers 2012; Mayer 2025) to generalise, and is likely to lead to restructuring. The research question here, then, is whether the opaque Catalan pattern is productive, and to the extent that it is not, what sort of “errors” (forms not conforming to the existing language pattern) might be produced by speakers when tested on novel stems. Such cases might take the form of just not applying /nt/-cluster reduction (i.e. /sant/ → ['sant]), or perhaps shifting (as Kiparsky originally suggested) to the transparent outcome, which would be ['sa], derived in rule-based phonology with feeding order: /nt/ → [n] → ∅.

2.3.2 Saltation and the /ɜ/ ~ [tʃ] alternation

Catalan has final devoicing: voiced obstruents at the end of stems are rendered as voiceless when the stem is followed by no suffix, thus placing the obstruent in final position. We give the rule with illustrative derivations in (9).

⁴ Certain highly unassimilated borrowings, such as (*Microsoft*) *Paint* and *PowerPoint*, can be pronounced with final [nt] (Pons-Moll 2015); these do not appear in our Wiktionary source.

(9) a. *Final devoicing*

$[-\text{sonorant}] \rightarrow [-\text{voice}] / ______]_{\text{word}}$ ‘Devoice obstruents finally’

b. *Derivations*

‘grey.M.SG’	‘grey-F.SG’	‘fat.M.SG’	‘fat-F.SG’	
/griz/	/griz-ə/	/gras/	/gras-/	underlying representations
s	—	—	—	final devoicing
[ˈgrɪs]	[ˈgrɪz-ə]	[ˈgras]	[ˈgrasə]	surface representations

The case of interest here concerns a set of masculine-feminine paradigms where [ɜ] alternates with [tʃ], as in [ˈbɔɜ-ə] ~ [ˈbɔtʃ] (*[ˈbɔʃ]) ‘crazy-F.SG/M.SG’. Following Bonet & Lloret (1998) and Wheeler (2005), we take /ɜ/ be the underlying form. Given final devoicing, we would expect /ɜ/ to surface as [ʃ], not [tʃ]. The appearance of [tʃ] is *not* the result of any ban on final [ʃ], for there also exist non-alternating /ʃ/ stems, such as [ˈfluʃ-ə] ~ [ˈfluʃ] ‘loose-F.SG/M.SG.’ In rule-based phonology, one option is to add a special version of the final devoicing rule, applicable only to /ɜ/, that also changes continuancy, as in (10a). As (10b) shows, it must be applied before ordinary final devoicing.⁵

(10) a. */ɜ/-to-[tʃ] Rule*

$\text{ɜ} \rightarrow \left[\begin{array}{l} -\text{voice} \\ -\text{continuant} \end{array} \right] ______]_{\text{word}}$ ‘Replace /ɜ/ by [tʃ] finally’

b. *Derivations*

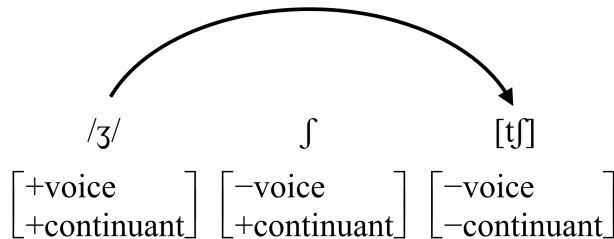
‘crazy-F.SG’	‘crazy.M.SG’	‘grey.M.SG’	‘loose.M.SG’	
/ˈbɔɜ-ə/	/ˈbɔɜ/	/ˈgrɪz/	/ˈfluʃ/	underlying representations
—	tʃ	—	—	/ɜ/-to-[tʃ] Rule
—	—	s	—	final devoicing
[ˈbɔɜə]	[ˈbɔtʃ]	[ˈgrɪs]	[ˈfluʃ]	surface representations

The shift of /ɜ/ to [tʃ] is an example of what Hayes & White (2015) call **saltation**, a term based on the Latin word for leaping. The underlying /ɜ/ of /bɔɜ/ surfaces as [tʃ], even though /ʃ/

⁵ There are alternatives. Instead of /ɜ/-to-[tʃ] we could have /ɜ/ to [dʒ], with the derivation completed by final devoicing. More fundamentally, Torres-Tamarit (2016) and Bonet & Lloret (2018) suggest that the underlying form for stems like [bɔɜ-ə] ~ [bɔtʃ] should be /bɔdɜ/, with [ɜ] derived by a rule of intervocalic spirantisation and [tʃ] by final devoicing. This analysis would require further apparatus, such as stem structure constraints, to explain why Catalan has no [ɜ] ~ [ʃ] alternations. In other words, to match the Catalan data pattern, the generalisation must be enforced that all stem-final phonetic [ɜ]s get derived from /dɜ/ rather than /ɜ/. If this alternative should prove correct, then a different theoretical question must be addressed, i.e. how language learners apprehend a principle like “all Xs are derived from Y”, perhaps itself a rather marked phenomenon; see McCarthy (2005) for discussion.

(phonetically intermediate between [ʒ] and [tʃ]) is phonotactically legal in this position; thus the /ʒ/ leaps across /f/, as illustrated in (11).

(11) *Saltation in Catalan*



Since the work of Łubowicz (2002) and Ito & Mester (2003), saltation, like opacity, has been known to be underivable in “classical” Optimality Theory, defined as Prince & Smolensky (1993) with the Faithfulness constraints of McCarthy & Prince (1995). Saltation *is* derivable in OT with various non-standard Faithfulness constraints; for example, Hayes & White use the *MAP constraints of Zuraw (2007, 2013).

Hayes & White (2015) also suggest that saltation is, in some sense, a marked phenomenon. It evidently only arises through quirky sequences of historical change, and is never innovated *per se* as a phonological pattern.⁶ Moreover, saltation appears to be difficult to learn, as is shown both by cases of historical restructuring—“saltation repair”—given by Hayes & White (2015), as well as White’s experimental results in artificial grammar learning (2014, 2017). Lastly, a principled explanation of the marked status of saltation is available, based on principles of the “P-map”, laid out by Steriade (2009) and developed formally by Zuraw (2013); deriving saltation requires constraint rankings or weightings that ban short phonetic paths like [ʒ] → [f] more strictly than long paths like [ʒ] → [tʃ].

The research question here is whether the productivity of the /ʒ/ → [tʃ] process falls short of the lexical pattern, which is indeed exceptionless (about 15 stems, including some outside our database,⁷ undergo /ʒ/ → [tʃ], and there are no stems at all with [ʒ] ~ [f] alternation). Our expectation is that if the participants do “repair” saltation, they will do so in the natural way, devoicing final /ʒ/ to the phonetically-close [f] rather than to phonetically-distant [tʃ]. Such forms, which mismatch the language pattern, may be interpreted as possible support for the view that saltation is hard to learn.

⁶ Indeed, this is so for the Catalan saltation, which arose from the diachronic events that are synchronically recapitulated in the analysis of Bonet & Lloret (2018: 229-230), cf. Moll (1952).

⁷ There are 7 stems in our lexical database; our figure of 15 reflects additional forms listed in Wheeler (2005:12, 260).

3. Experimental study

To review, our study of phonological productivity focuses on two related questions: (1) whether processes that have many exceptions can be productive, and if so, whether the details of such processes are frequency-matched; (2) whether processes often considered to be unnatural, because they are involved in an opaque interaction or are saltatory, suffer from lessened productivity. To this end, we conducted two experiments. In Experiment 1, participants produced their favoured wug responses orally, which were then transcribed by the authors in IPA. In Experiment 2, we asked the participants to rate on a 1-7 scale the responses given most frequently by the participants in Experiment 1. Including this receptive task served two purposes. First, it allowed us to test whether the patterns observed in production would be replicated in a different modality. Second, it enabled us to determine whether participants would also accept alternative candidates for a particular process, even when production data suggested a clear preference for a particular candidate.

3.1. Participants

For Experiment 1, 43 participants were recruited. Six were excluded, for various reasons: their first exposure to Catalan occurred after age 3 ($n = 2$),⁸ they self-identified as non-Central Catalan speakers ($n = 2$), or they failed a training task item (see §3.3; $n = 2$). This left 37 participants. For Experiment 2, 51 participants were recruited. Fourteen were excluded due to late age of acquisition ($n = 4$), self-identifying as non-Central Catalan speakers ($n = 7$), and/or failing at least one of the control items (§3.3, $n = 3$), leaving 37 participants.⁹

In a post-experiment questionnaire, all included participants reported that they spoke Catalan at home during childhood, and the vast majority had completed their mandatory education (ages 5-16) in Catalan. We note that our participants were strongly multilingual—the median rate of Catalan usage was only 60%—and that this may have influenced our results (see §4.2.1, §4.4). It would be logically possible to avoid such effects by recruiting a participant population consisting of monolinguals, but this seems unrealistic given that Catalonia is a highly multilingual society.¹⁰

⁸ For evidence that degree of language exposure in Catalan strongly influences the propensity to alternate, see Jovanovich-Trakál (2021).

⁹ As it happens, eight speakers elected to participate in both experiments; this was not part of the experimental design, which was not intended to identify within-subject effects. None of the eight participants who did both experiments saw the same wug items. To ensure that there were no group differences between those that participated in both experiments and those that did not, we reran the statistical models for Experiment 2 with the inclusion of a factor `RepeatedParticipant` for those participants who participated in Experiment 1. There was no significant effect of `RepeatedParticipant`.

¹⁰ According to the Statistical Institute of Catalonia, in 2023, 43.4% of Catalans aged 15 years or older speak English at a conversational level (<https://www.idescat.cat/indicadors/?id=basics&n=10367&tema=cultu>). For Spanish, the percentage is 99.2%.

Participants were recruited online with the assistance of government agencies and nonprofit organisations that promote the Catalan language; we also found some participants through word of mouth. Participants were compensated for their time with a USD\$15 electronic gift card.

3.2 Materials and Design

3.2.1 Stimuli

For all stimuli, the participants were provided with feminine forms and asked to produce the corresponding masculines. The rationale is that the feminine forms, where the relevant phonology is inapplicable, provide the participants with the information needed to set up the underlying representation of the wug stem.

The stimuli contained four different conditions, as shown in (13). There were two conditions for each of our two research questions. The stimuli for /n/-deletion and /r/-deletion involve four subconditions. The rightmost column of (13) gives the choices that were offered to participants in Experiment 2, which were drawn from the most frequent responses given in Experiment 1.

(13) *Experimental conditions and subconditions*

<i>Condition</i>	<i>Subcondition</i>	<i>Presented to participants in Expts. 1 and 2</i>	<i>Options for participants in Expt. 2</i>
Exceptionful phonology	frequent suffix (/inə/, /dɔrə/)	/n/: [bəlun'trin-ə] /r/: [gruə'dɔr-ə]	[bəlun'tr-i], [bəlun'trin] [gruə'do], [gruə'dɔr]
	monosyllabic	/n/: ['frun-ə] /r/: ['lɛr-ə]	['fru], ['frun] ['lɛ], ['lɛr]
	penultimately stressed	/n/: ['dɔstun-ə] /r/: ['sɔlir-ə]	['dɔstu], ['dɔstun] ['sɔli], ['sɔlir]
	other	/n/: [gə'mɛn-ə] /r/: [kə'nar-ə]	[gə'mɛ], [gə'mɛn] [kə'na], [kə'nar]
Unnatural phonology	Opacity: /nt/-cluster simplification	[mir'bunt-ə]	[mir'bun], [mirbunt], [mir'bu] (feeding order)
	Saltation: /ʒ/ final obstruent devoicing	[sə'lɔʒ-ə]	[sə'lɔʃ], [sə'lɔʃ]

We characterise the stimuli in more detail below.

Exceptionful phonology: We chose our wug items to test the general hypothesis that speakers will frequency-match in cases of exceptional phonology. There were four environments, each reflecting how deletion rates vary in the lexicon.

- a) **Frequent suffixes.** 20 polysyllabic stems, 10 with the suffix /-inə/ and 10 with the suffix /-dora/. We expected high rates of deletion given the unanimous pattern found in the lexicon.
- b) **Monosyllabic stems.** 20 monosyllabic stems, 10 ending in /n/ and 10 ending in /r/. We expected fewer cases of deletion, both because this matches the lexicon and because of the possible learning bias against alternation in initial syllables (Becker *et al.* 2012).
- c) **Penultimately-stressed stems.** 20 polysyllabic stems with stem-penultimate stress, 10 ending in /n/ and 10 ending in /r/. We expected lower rates of deletion given the lexical pattern.
- d) **Other.** 20 polysyllabic stems with stem-final stress and not ending in *-ina* or *-dora*: 10 ending in /n/ and 10 ending in /r/. These were chosen as a sort of baseline, assessing preference for deletion in the absence of any of the first three environments.

Unnatural phonology: We chose our wug items to investigate the rates of productivity among the two “unnatural” alternations which are exceptionless in the lexicon (see §2.3; research question (2b)).

- a) **/nt/-final cluster simplification:** 10 polysyllabic stems, with stem-final stress, no identifiable suffix, and ending in /nt/. These were intended to test the productivity of the opaque interaction of /nt/-cluster simplification and /n/-deletion (research question (2bi)).
- b) **/ʒ/ final obstruent devoicing:** 10 polysyllabic stems, with stem-final stress, no identifiable suffix, and appearing with [ʒ] in the feminine. These were intended as a test of the productivity of saltation (research question (2bii)).

In total, 100 different wug forms were created across 10 subconditions.¹¹ The complete list can be found in the Supplementary Materials (available from the editorial platform or from [OSF](#)).

Wug items were found by using a script to generate a very large set of phonotactically legal candidates. The wug items were chosen from this list by the native speaker author, with the goals of maximizing phonological naturalness as well as variety. All items were checked to make sure that neither they nor the anticipated participant responses to them were existing words of Catalan.¹²

¹¹ The experiment also included an additional condition, with 30 items embedded in six frames beyond the 20 frames discussed below, which tested a separate research question, the ability of speakers to project URs from surface forms (Ernestus & Baayen 2003). We do not discuss this condition here for reasons of space. The data for these items may be obtained from the OSF site.

¹² After piloting and completing data collection for Experiment 1, we detected two items that could be or were in fact existing low-frequency words: [mu.ɲi.'do.rə] could have been interpreted as ‘milk’, derived from *munyir* ‘to milk (V)’; and one of the candidates for ['klɔ.rə], ['klɔr], means ‘chlorine’. Neither item elicited responses that

3.2.2 Frame paragraphs

We created 20 frame paragraphs, which were used in both experiments; an example is given in (14). The frames were presented auditorily and in written form on the screen, but the wug words were only presented auditorily; on the screen, they appeared as blanks. The frames were designed to give the participants multiple exposures to the wug word before they actually rendered the crucial wug-test response. The frame paragraphs are listed in the Supplementary Materials.

(14) *Sample frame paragraph ([bə.lun. 'tri.nə] as adjective)*

[bə.lun. 'tri.nə]₁

Una obra [bə.lun. 'tri.nə]₂ era una peça d'art on s'havien aplicat tècniques mixtes amb ornaments de metalls i pedres precioses. Al segle XV, un artista català va crear la primera escultura ____₃, feta de marbre, pedres precioses, i or. El primer quadre ____₄ no es va crear a Espanya fins al segle XVII.

'[bə.lun. 'tri.nə]₁

'A [bə.lun. 'tri.nə]₂ work was a piece of art where they had applied mixed media with precious metals and stone ornaments. In the 15th century, a Catalan artist created the first ____₃ sculpture, made of marble, precious stones and gold. The first ____₄ painting was not created in Spain until the 17th century.'

The frames were constructed to discourage participants from treating them as foreign words, which is particularly important in light of the fact that /n/-deletion, /r/-deletion and /nt/-cluster simplification are underapplied in Catalan loanwords (Pons-Moll 2015, 2021). For this reason, we crafted the content of the paragraphs to encourage participants to treat the wug forms as antiquated words in Catalan that had gone out of use, rather than loanwords.

The key task for participants was to internalise the wug form as much as possible, then provide a response inflecting it in the masculine (Experiment 1) or to rate a number of possible candidates for the masculine (Experiment 2).

Participants first heard the wug word in isolation (position 1), then in a sentence (position 2). In Experiment 1, position 3 was left as a gap; it required participants to repeat the wug word to check that they had internalised it correctly ($n = 41/740$ trials with incorrect repetitions were excluded). In Experiment 2, where no recording was collected, position 3 was simply pronounced again as part of the frame. Position 4 implemented the actual wug test: participants

diverged from the items of their respective subconditions: we obtained 100% deletion for [mu.ni. 'do.rə] (/dot/ subcondition average = 100%) and 33.33% deletion for ['klɔ.rə] (monosyllabic r-deletion average = 26%).

were required to speak their response in Experiment 1 and choose from among options ((13) above) for Experiment 2.

There were a total of 100 wug words and 20 frame paragraphs, hence five different words per frame. We paired each wug word with a single frame in pseudo-random fashion, ensuring that the wug words ending in *-dora* and *-ina* were paired with nouns and adjectives, respectively. Other than those wug items, half of the remaining wug words were randomly placed in nominal contexts, the others in adjectival contexts. Each participant heard two stems each from our 10 subconditions (13).

The 100 frame + wug combinations were recorded by the second author in a sound booth using Praat (Boersma & Weenink 2019) with a mono audio channel, 44.1 kHz sampling rate, and 16 bits per sample. The recording was made in a way meant to ensure prosodic realism and maximum uniformity. Thus, for (14), the sentence containing the wug was recorded separately for each of the five wug forms affiliated with this frame, then spliced to a single recording of the remainder of the frame. Each gap was edited to be 1.5s long. All recordings were modified to have 72 dB of intensity. The audio files are available in the Supplementary Materials.

3.3. Procedure

The experiments were created and distributed using Labvanced (Finger *et al.* 2017). Participants received a link to the experiment and participated online using their own computer.

3.3.1 Experiment 1: Production task

Experiment 1 was an elicitation task, in which participants freely provided whatever response to the experimental material they felt appropriate given the linguistic context. For each of our subconditions, each participant completed two trials; hence we analyse a total of 20 test trials per participant. An example trial, with discussion of the procedure followed, is given in §3.2.2. Before the start of the experiment, each participant saw two practice trials using similar paragraphs, one with a real masculine word ([bu.'ta.nik] 'botanist.M.SG'), and one with a feminine wug word that did not exhibit any alternation of interest, ['grɛ.zə]. Participants were required to inflect the real word as expected in order to be included in the study. Participants were not allowed to skip trials. Their responses were timed (from the end of the sentence frame to the click). The experiment took the participants on average 44 minutes.

After the experiment, participants were asked a number of demographic and linguistic questions. Some were intended to ensure that the included participants were exposed to Catalan during infancy, spoke Catalan at home, and self-identified as Central Catalan speakers. We additionally asked their age, gender, educational level, and how often they speak in other languages during a typical week. Lastly, we asked participants to rate from 1 to 7 whether their responses throughout

the experiment were “1 = purely intuitive” or “7 = based on conscious reasoning,” based on Moreton & Pertsova’s (2023) suggestion that participants return different answers to wug test questions when they reason through their response consciously rather than relying on intuition. It turned out that these factors played only a minor role in predicting participant responses; for discussion see §4.4.

3.3.2 Experiment 2: Acceptability judgement task

Experiment 2 was an acceptability judgment task, in which participants were given two or three prerecorded possible phonological outcomes that might appropriately appear in the frame paragraph, and rated each one on a Likert scale from 1 to 7. The materials were identical to Experiment 1, except that the first two gaps in the frame paragraph contained the prerecorded wug word while the last gap provided the context for the forms that the participants were asked to evaluate. For the choices made available to the participants and the research questions addressed, see §3.2.1.

Before the start of the experiment, each participant underwent three practice trials. For the real word *botànic* ‘botanist.M.SG’ they rated a correct inflected form ([bu.‘ta.ni.kə]) and an incorrect one ([bu.‘ta.ni.ə]). For the wug word [‘grɛ.zə] they were given the (very likely) choice [‘grɛs] and an unlikely choice [‘grɛ]. There was also a practice trial with the feminine wug form [əz.‘ma.βə], which, following Catalan phonology (Wheeler 2005:§10.3) has two acceptable masculine forms, namely [əz.‘map] or [əz.‘maw]. The purpose was to ensure participants understood that they could accept multiple candidates by rating them all high, or reject multiple candidates by rating them all low. The practice trials were accompanied by hints about how to rate these words (e.g., “for option (a), [‘grɛ], most Catalan speakers would assign a low score (1 or 2)”, for [əz.‘ma.βə]) “sometimes, there may be more than one plausible answer”).

There were four control items, embedded in similar paragraphs. These were interspersed with the test items and contained real words with only one possible correct answer. Two were masculine forms (*catedràtic* ‘professor.M.SG’, *filòsof* ‘philosopher.M.SG’) and two were feminine (*analítica* ‘analytical.F.SG’, *autònoma*, ‘autonomous.F.SG’). None of them exhibited the alternations of interest. Participants had to rate the correct candidate higher than the incorrect candidate in order to be included in the study. For example, for *analítica* the candidates were [ə.nə.‘li.tik] (correct) vs. [ə.nə.‘li.ti] (incorrect). In total, participants had to evaluate six training items (within three trials), 42 test items (within 20 trials), and eight control items (within four trials). The experiment took on average 21 minutes.

After completing Experiment 2, participants were asked to complete the same questionnaire used in Experiment 1 (§3.3.1).

4. Results and Discussion

For Experiment 1, the recordings were transcribed by two phonetically-trained listeners with disagreements adjudicated by a third native-speaker transcriber. Forty-six tokens were excluded due to the participants incorrectly repeating the wug form (blank 3 of frame (14)). This left 699 tokens to be included in the analysis. For Experiment 2, we report raw rating scores.¹³ In total, 1860 ratings were reported across the various wug items and candidates. Full results for both experiments, including individual participant responses and the R scripts for statistical analysis, are available in the Supplementary Materials.

Given the similarity of the results, we report the two experiments together.

A note on the data reporting: in the course of our study, we found that several participants provided unexpected responses. For example, several participants used the low frequency masculine suffixes [-u] and [-ə] to “dodge the question” of whether a particular process should apply. This is because the addition of a suffix to a stem eliminates the environment for the various phonological processes under investigation to apply; we will call these “avoidant” responses. We also encountered some responses that make no sense in light of the grammatical patterning of Catalan, such as [ʎu'da] for [ʎu'daʒə]; we call these “aberrant” responses. In computing totals below, we exclude both avoidant and aberrant responses; we discuss them in more detail in §4.3.

4.1 Productivity of exceptional phonology

The first point that emerges from the data is that /n/- and /ɾ/-deletion are productive, in that the participants frequently generalise deletion to novel items—although to different degrees: the overall deletion rate was 80% for /n/ and 52% for /ɾ/. The observed degree of productivity is of interest because previous work has suggested that these processes are unproductive (Wheeler 2005, Pons-Moll 2015). The high application rate we obtained is perhaps to be attributed to our efforts to characterise the wug forms as unknown native words, thus overcoming the tendency to treat loanwords faithfully.

As shown in Figures 1 and 2, the rates of deletion differed substantially across subconditions.

¹³ We also performed the analyses using Z-score normalised ratings; however, the statistical patterns were the same.

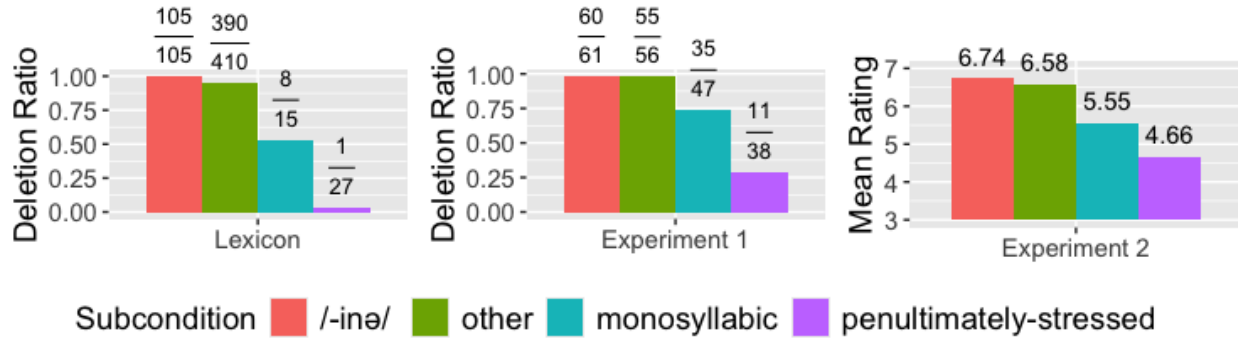


Figure 1. /n/-deletion in the lexicon, Experiment 1 (production), and Experiment 2 (ratings)

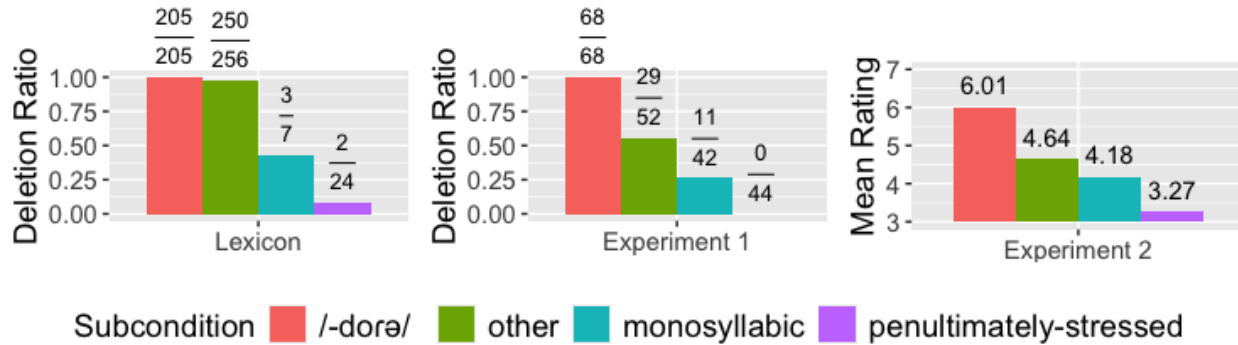


Figure 2. /r/-deletion in the lexicon, Experiment 1 (production), and Experiment 2 (ratings)

Informal inspection of Figures 1 and 2 suggests there is indeed frequency-matching of the Lexicon: Subconditions with higher application rates in the Lexicon tend to have higher application rates in Experiment 1, with high ratings in Experiment 2. To test these claims statistically, we constructed two models, both of which combined the /n/ and /r/ data. For Experiment 1, we used a mixed-effects logistic regression model, in which the binary dependent variable was the response given (deletion = 1, nondeletion = 0) for each combination of participant and wug word. We included LexicalRate (from leftmost graphs in Figures 1 and 2) as a fixed effect and a random intercept for Subject using the lme4 package in R (Bates *et al.* 2015).¹⁴ For Experiment 2, a cumulative-link mixed-effects model was employed using the ordinal package in R (Christensen 2025). This model included rating (1-7) as the dependent variable, LexicalRate as a fixed effect, a random intercept for Subject, and random slopes for Wug Item by Subject.

It emerged that there is a significant effect of LexicalRate in both Experiment 1 ($\beta = 0.83$, $SE = 0.05$, $t = 17.29$, $p < 0.001$) and Experiment 2 ($\beta = 2.73$, $SE = 0.37$, $z = 7.45$, $p < 0.001$),

¹⁴ Models with more complex random effect structures (random slopes and interactions) did not converge. Including a term for lexical attestation (number of cases in lexical database) did not improve the fit of the model (see OSF site).

confirming that the rate at which a process applies in the lexicon significantly predicts the application rate in Experiment 1 and ratings in Experiment 2.¹⁵

An alternative explanation for frequency-matching is that it reflects idiolect mixture, i.e. a statistical blend of responses from a mixed population of participants each of whom has one unique preference (Halle & Vergnaud 1981). Our data argue against this. Experiment 1 gave each participant two items from each condition, and many participants gave different responses to these two items. The rate of matching in such pairs was only 67%, far below the 100% this view predicts and close to chance expectations, which we calculated at 72%, obtained by sampling responses at random from the observed probability distribution. Experiment 2 similarly suggests that gradience exists within individual speakers, as ratings are often clustered in the middle of the scale rather than at both extremes.

Our results on /n/-deletion replicate and extend those of the only previous wug-testing work on Catalan known to us, namely Jovanovich-Trakál (2021), who found modest productivity for /n/-deletion in plural alternations like [klə'fon-s] ~ [klə'fo(n)] ‘wug.M-PL/M.SG’. Our own participants showed higher productivity for /n/-deletion. We suggest that this may arise from (a) our use of spoken language rather than orthography in the presentation of the stimuli; (b) our more stringent criterion for Catalan exposure in childhood; (c) the fact that we tested adults and Jovanovich-Trakál (2021) children.

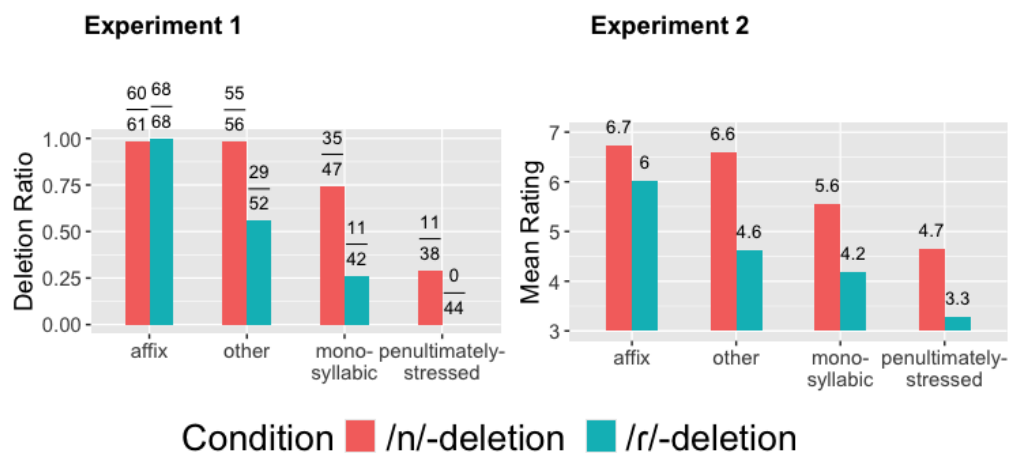


Figure 3: /n/- and /r/-deletion in Experiment 1 (production) and Experiment 2 (ratings)

There is a second pattern that emerges from visual comparison of Figures 1 and 2: all else being equal, the rate of /n/ deletion appears higher than that of /r/ deletion; and the same holds for the rating data of Experiment 2. It seems that /n/-deletion closely matched the lexical frequencies but

¹⁵ For statistical testing of pairwise comparisons across subconditions of the rate of deletion in Experiment 1 and the ratings in Experiment 2, see the Supplementary Materials

/r/-deletion matched them only in *relative* terms. The comparison emerges more clearly when we directly juxtapose the two data sets, as in Figure 3.¹⁶

To confirm this effect statistically, we ran augmented versions of the logistic regression and cumulative-link mixed effects models described, adding in a fixed effect of Consonant (/n/ vs. /r/). Since the paired models are nested, we performed a Likelihood Ratio Test on each pair. The test indicated that Consonant mattered at a significant level for both Experiment 1 ($\chi^2(1) = 57.33$, $p < 0.001$) and Experiment 2 ($\chi^2(1) = 106.05$, $p < 0.001$).¹⁷

We offer here speculation for why /n/-deletion applies at a higher rate, and is rated better, than /r/-deletion, despite their similar lexical profiles. One possible explanation is dialect variation, described in Wheeler (2005). Speakers of Central Catalan encounter speakers of another major dialect, Valencian, which lacks /r/-deletion; whereas /n/-deletion is pan-dialectal. Another possibility is that participants were influenced by orthography, an effect observed in earlier research (Daland, Oh & Kim 2015; Kawahara 2018). In Catalan, /n/-deletion is spelt out: ['san-ə] ~ ['sa] is spelt *sana* ~ *sa*, but /r/-deletion is not: ['klar-ə] ~ ['kla] is spelt *clara* ~ *clar*. Our participants may have been constructing appropriate orthographic representations for what they heard, preferring to pronounce these representations faithfully; for further discussion see §5.2 below.¹⁸

4.2 Productivity of unnatural phonology

4.2.1 Opacity results and discussion

Our results show that /nt/-cluster simplification is productive: a majority of responses (52%) in Experiment 1 were of the form /nt/ → [n], Figure 4. However, [nt] answers were frequent: these formed 42% of the responses in Experiment 1, and in Experiment 2, [nt] was actually rated numerically higher than the expected [n]; see Figure 4.

¹⁶ The difference bears on a methodological issue raised by one of our reviewers, namely why one might want to test more than one phonological process in the same experiment. Here, the fact that /n/-deletion gave responses that closely matched the lexicon shows that the lower rates for /r/-deletion are unlikely to result from some problem in the experimental design; we could not have known this had we tested /r/-deletion alone.

¹⁷ Although /n/ and /r/ behave differently, it remains the case that there is frequency-matching for both consonants in both experiments. We determined this by subsetting the /n/ and /r/ data and rerunning the models reported above (Experiment 1, /n/: $\beta = 6.54$, $SE = 1.24$, $z = 5.29$, $p < 0.001$. /r/: $\beta = 5.31$, $SE = 0.66$, $z = 8.06$, $p < 0.001$; Experiment 2, /n/: $\beta = 3.69$, $SE = 0.39$, $z = 9.51$, $p < 0.001$. /r/: $\beta = 2.52$, $SE = 0.31$, $z = 8.27$, $p < 0.001$).

¹⁸ Yet another explanation for the /n/ vs. /r/ difference is that our stimuli inadvertently included items in which the outcome could be influenced by having two /n/s or two /r/s in the same word; a so-called “OCP” effect (McCarthy 1988 et seq.). Our statistical testing suggest a significant effect of OCP, in the single context [rVr], but the difference goes in the opposite direction, i.e., an OCP effect would predict more /r/-deletion than /n/-deletion, and the contrary is true in our data.

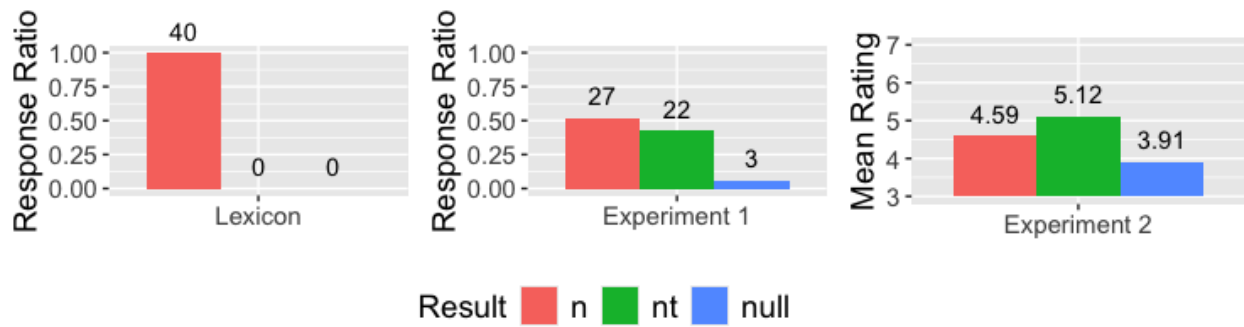


Figure 4: /nt/-cluster simplification in the lexicon, Expt. 1 (production), and Expt. 2 (ratings)

The frequency with which /nt/ is retained is surprising because the phenomenon is exceptionless in the lexicon, and because /nt/ simplification can plausibly be regarded as more natural than /n/- or /t/-deletion (see Pons-Moll 2015): it is typologically more common (Bonet *et al.* 2005), and imposes a lesser Faithfulness cost in terms of perceptual salience (Steriade 2009). Indeed, in comparing participants' rates of [nt] responses to the lexical frequencies, we find that [nt] responses are significantly overproduced in Experiment 1 relative to the (zero) lexical frequency (*Fisher's exact test*, $p < 0.01$). We offer some conjectures for this asymmetry below.

The original reason we included /nt/-cluster simplification in the experiment was to test whether opaque phonology (here, counterfeeding of /n/-deletion) can be productive. To see how our results bear on this question, we examined the proportion of participants that fell into the patterns outlined in Table (15). In calculating the proportions in the table, we omitted participants who did not provide enough relevant responses (non-avoidant, non-aberrant) to make comparison possible; we also omitted cases where the participant gave different responses for either the /nt/ pair or the /n/ pair.

(15) Participant response patterns for /nt/-cluster simplification and /n/-other deletion.

<i>Pattern</i>	<i>Fraction of participants in Expt. 1</i>	<i>Fraction of participants in Expt. 2</i>
a. <i>Opaque</i> nt → n, n → ∅	13 (46%)	7 (32%)
b. <i>Suppress /nt/ simplification</i> nt → nt, n → ∅	13 (46%)	12 (55%)
c. <i>Feeding</i> nt → ∅, n → ∅	1 (4%)	3 (14%)
d. <i>Delete /nt/</i> nt → ∅, n → n	1 (4%)	0 (0%)
e. <i>Suppress /n/-deletion</i> nt → n, n → n	0 (0%)	0 (0%)
f. <i>Fully faithful</i> nt → nt, n → n	0 (0%)	0 (0%)
g. <i>Inconsistent</i>	2	13
h. <i>Other</i> ¹⁹	7	2

The Experiment 1 data suggest that counterfeeding opacity (15a) can be productive: it is found for 46% of the participants. This constitutes evidence that very strong positions stating that opacity is unproductive, such as Sanders (2003), are likely to be untenable.

On the other hand, many other responses deviated from the (unanimous) lexical pattern, in a way that might be interpreted as “opacity repair”. These cases fall into different categories. Thirteen participants (46%) gave responses of type (15b): [mir'buntə] → [mir'bunt], [gə'mən-ə] → [gə'mɛ]. This is opacity repair in the sense that if you do not apply /nt/ deletion, the resulting output keeps /n/-deletion transparent. Only a single participant volunteered responses of type (15c): [mir'buntə] → [mir'bu], [gə'mən-ə] → [gə'mɛ]. This pattern matches the prediction made in Kiparsky's (1968) original work on opacity, which asserted that the feeding order (here, /nt/ → n → [∅]) is unmarked. However, this outcome occurred so rarely that we are tempted to classify

¹⁹ This represents cases where the participant only provided avoidant or aberrant responses which did not allow for comparison.

it as an aberrant response, similar to [λu'daʒə] → [λu'da]. Therefore, the results do not support the original Kiparskyan prediction.

In sum, we observed substantial numbers of cases where the opaque lexical pattern was directly reflected in the participant's responses, roughly equal cases where transparency was achieved through suppression of the feeding process, and very few cases of any other type (including the case of Kiparskyan feeding order). The results are compatible with the view that opacity is somewhat difficult, but not impossible, to learn (Ettlinger 2008; Kim 2012; Jarosz 2015; Prickett 2019).

The Experiment 2 data roughly match those of Experiment 1, though with slightly lower acceptance of the opaque pattern and a higher level of inconsistency.

Lastly, we acknowledge there are other possible factors that may have influenced the unexpectedly low rates of /nt/-cluster simplification and the relatively high production and ratings of [nt] forms: (i) Exposure to other languages or Catalan dialects that do not have /nt/-cluster simplification (Wheeler 2005:221) may weaken the native-language phonotactic constraint banning final [nt];²⁰ for a possible case of L2 learning leading to “weaker” phonology in L1 see Dmitrieva *et al.* (2010); (ii) Orthographic influence, as in the case of /r/-deletion, since /nt/-cluster simplification is not spelt out.

4.2.2 Saltation results and discussion

We included the saltatory alternation [ʒ] ~ [tʃ] to test hypothesis (2bii), that saltation is unstable and vulnerable to restructuring. Our findings are summarised in Figure 5.

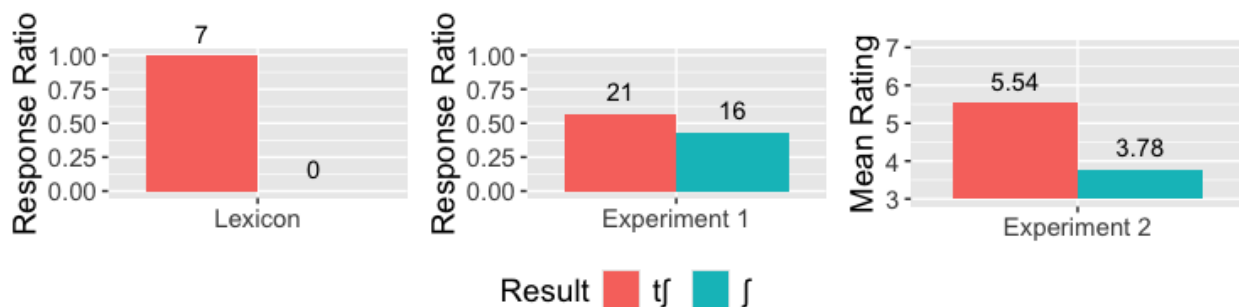


Figure 5: [ʒ] ~ [tʃ] alternation in the lexicon, Expt. 1 (production), and Expt. 2 (ratings)

To start, saltation does seem to be productive in the sense that the majority of responses (57% or 21/37) took the saltating pattern [λu'daʒə] → [λu'datʃ]. However, in comparing participants' saltation rates to the lexical frequencies, we find that saltation is significantly underproduced in

²⁰ This might predict that self-reported English language exposure would be related to frequency of non-application for /nt/-cluster simplification; but we found no significant effect; see §4.4 below.

Experiment 1 relative to the lexical frequencies (*Fisher's exact test*, $p = 0.04$). Many speakers produced forms that repaired saltation, like [ʎu'daʒə] → [ʎu'daʃ]; this was 43% or 16/37 of the phonological responses.²¹ The saltation repair occurred despite the fact that forms with [ʒ] ~ [ʃ] are not attested in the lexicon nor in any other dialect of Catalan. The observations are thus compatible with the view that saltation, as an unnatural process, goes underlearned in Catalan.

As before, we offer speculation concerning the observed lexicon-experiment mismatch. The lower-than-expected rates of saltation cannot be accounted for by orthography (as we suggested above for opacity), because the saltation alternation is spelt. Further, it is unlikely that the lower rates could be the result of dialect differences, for the saltation pattern is found in all common dialects. However, another factor may have contributed to the low production of saltation. In particular, the participant responses may simply have reflected the small number of forms in the lexicon that bear on the question (7 in our database, §2.3.2, approximately 15 in total in the language if we include verbs).

In Experiment 2, the [ʃ] outcomes were rated quite low compared to how often they were produced in Experiment 1. We can compare the two simply by sorting the Experiment 2 participants by which candidate they preferred. Excluding responses where equal ratings for both candidates were given ($n = 7$), 75% (50/67) of responses showed a preference for the saltatory outcome compared to 57% (21/37) for Experiment 1. A possible reason for the difference is that in Experiment 1, participants did not readily access the [tʃ] candidate, since this alternation is not well attested in Catalan; thus, they responded with [ʃ] because it appeared to be the only plausible option. In contrast, in Experiment 2, [tʃ] was given as an explicit option, so there was no difficulty in assessing it as a viable candidate.

In summary, the results from the opacity and saltation conditions indicate that unnatural phonological processes are productive, but not entirely so. In both cases, participants underapplied the process relative to the lexicon, which exhibits categorical (100%) rule application. This pattern is consistent with the hypothesis that unnatural phonological processes can lead speakers to deviate from frequency-matching.

4.3 Non-phonological responses

While our study was focused on outcomes related to phonology, we will cover here some results that we found interesting but did not fit into the preconceived scheme of our experiments.

²¹ Note that the simplest saltation repair would have been non-alternating [ʎu'daʒ]. We suggest that this outcome never arose because final devoicing remains a powerful phonotactic principle of the language; for instance, unlike any of the other processes discussed here, final devoicing is a characteristic of Catalan-accented L2 speech (Pons-Moll 2015).

Notably, some participants employed morphology in unexpected ways. As observed in §2.1, Catalan has two forms of irregular masculine inflection. The suffix [-u] is found in examples like [bu'ra.tʃ-**u**] ~ [bu'ra.tʃ-ə] ‘drunk-M.SG/F.SG’. It occurs in 0.95% (55/5761) of the paradigms in our database. The irregular masculine ending [-ə], seen in [ə'legr-ə] ~ [ə'legr-ə] ‘happy-M.SG/F.SG’, is found in 0.71% (41/5761) of paradigms in our database. In Experiment 1, we found that 2.1% (15/699) of participant responses employed [-u], a higher value than the percentage observed in the Catalan lexicon. For [-ə], such responses were even more frequent: 16.7% (117/699).

We offer a conjecture for why participants volunteered these morphologically unusual masculine forms so frequently: since the irregular endings provide a vowel after the stem, they allow a participant to circumvent the task of answering the phonological question at hand. Based on this possible explanation, we refer to responses with [-u] or [-ə] as **avoidant**. Avoidant responses in wug-testing have been noticed before, particularly among children. Do (2018) found abundant cases in which Korean children chose a morphologically somewhat inappropriate response that allowed them to bypass the choice of what form of phonological alternation to deploy. See also Zamuner, Kerkhoff and Fikkert (2012) and Kerkhoff (2004, 2007) for Dutch-speaking children, Pérez-Pereira (1989) for Spanish-speaking children, and Kim (2025) for Spanish-speaking adults.

The connection to the child language literature offers a possible account for why avoidance with [-ə] was much more common than avoidance with [-u], despite the statistics of the lexicon:²² children are commonly observed in wug-testing to engage in the behaviour of simply repeating the wug word verbatim (e.g., Pérez-Pereira 1989; Zamuner, Kerkhoff & Fikkert 2012); we may be observing a similar pattern among adult speakers (cf. Kim 2025).

We hypothesise that avoidance is a consequence of uncertainty; specifically, that speakers are reluctant to project a pattern when their lexicon includes relatively few examples. For instance, the highest avoidance rate, 34.7%, was seen for the /t/ monosyllabic subcondition, which had the lowest number of lexical examples ($n = 7$); and the lowest avoidance rate, 2.9%, was for the /-dɔrə/ subcondition, which is very well attested in the lexicon ($n = 205$). To test this hypothesis statistically, we ran a linear model with the `lm()` function in R. Log-transformed word frequency was regressed on the proportion of avoidant responses for each subcondition. The model revealed a significant negative relationship; that is, a low number of examples for a given subcondition in the lexicon predicts avoidant responses ($F(1,8) = 12.6, p = 0.008$). This finding suggests that native speakers are tacitly aware of cases where their response is based on few data

²² Note that in most of the masculine schwa forms, the schwa can be treated as the result of epenthesis, repairing illegal final sequences (e.g., /ə'legr/, → [ə'legrə] ‘happy.M.SG’; Wheeler 2005:§8.3); there are very few cases like [pu'et-ə] ‘poet.M.SG/F.SG’ or [pəriu'dist-ə] ‘journalist.M.SG/F.SG’, that cannot be derived by epenthesis (final [t] and [st] being phonotactically legal). None of the schwas in our wug test could be epenthetic. Therefore, the lexicon-wug test disparity is perhaps even larger than indicated above.

points, and in such cases they are reluctant to guess. We know of no phonological models that predict this finding and see it as a challenge for future research.

We also investigated whether there was a relationship between avoidance and response times (total time of trial – total time of recorded frame). To do this, we first screened the reaction times for outliers. Trials with response times greater than $1.5 \times$ the interquartile range above the third quartile or less than $1.5 \times$ the interquartile range below the first quartile were excluded from analysis. A linear mixed-effects model was then fitted to examine whether avoidant responses were associated with longer response times, with Avoidance (avoidant vs non-avoidant) as a fixed effect and random intercepts for Subject. The model showed a marginal effect of avoidance on response time ($F(1, 37) = 3.51, p = 0.062$), with avoidant responses tending to be slower than non-avoidant responses. The average response time was 25.0 s for avoidant responses vs. 21.9 s for non-avoidant responses. As with lexical frequency, the greater response times suggest greater uncertainty: perhaps they reflect a process whereby participants tacitly weighed multiple phonological options, felt unable to choose between them, and ultimately opted for a non-phonological solution.

Aside from avoidant responses, we observed a number of responses (9%, 62/699) that might be called **aberrant**, as they lacked any sort of reasonable interpretation. One instance occurred with [ʎu'daʒə], intended to test for saltation (§2.3.2): six participants responded with deletion, i.e. [ʎu'da]. Conceivably, this response, which has no precedent in the Catalan lexicon, was also avoidant; though bizarre, it lets the participant avoid choosing between [ʎu'datʃ] and [ʎu'daʃ]. Just as with the avoidant responses, we statistically tested whether there was a relationship between aberrant responses and response times. As with the previous model, we excluded outliers before fitting a linear mixed-effects model with Aberrance as a fixed effect and random intercepts for Subject. The model showed a significant effect of Aberrance on response times ($F(1, 36.4) = 15.17, p < 0.001$) with aberrant responses involving greater average response time (29.3 s for aberrant responses vs. 21.8 s for non-aberrant). We also used a linear model to test whether there was an effect of low lexical attestation on rates of aberrant responses. However, we did not find a significant effect ($F(1, 88) = 1.4646, p = 0.26$).

Since the aberrant responses do not respond to statistical frequency, we are inclined to regard them as noise, uninformative with respect to the participants' phonological knowledge.

4.4 Demographic and test-taking factors

As noted in §3.3.1, we asked our participants to answer questions regarding sex, level of education, language of education, knowledge of foreign languages, and whether they made their responses consciously or intuitively (Moreton & Pertsova 2023). To assess the effects of these factors, we carried out logistic regressions for each Condition (/n/-deletion, /r/-deletion, saltation,

/nt/-cluster simplification) which included factors for Subcondition (for /n/- and /r/-deletion only) along with age, university-level education in Catalan, overall level of education, knowledge of English, and whether they responded intuitively or consciously as fixed effects and participant as a random intercept. Across several models for each condition, none of these factors, with one single exception, had a significant effect on participant responses. The one factor that seems to matter is age: older participants are more willing to apply /r/-deletion ($\beta = 0.03$, $p = 0.02$). We conjecture, in light of Catalan history, that such speakers received less of their education in Catalan and that the effect of orthographic suppression of /r/-deletion, discussed in §4.1, is weaker for them.

5. Summary and conclusions

5.1 Summary of findings

To conclude, we return to the theoretical questions that motivated this study: the productivity of **exceptionful** and **unnatural phonology**.

Our first research question concerned the status of exceptionful phonology as exemplified by /n/- and /r/-deletion. We asked whether participants tested on novel words would productively extend exceptionful processes, and if so, whether they would frequency-match the patterns of the lexicon (Zuraw 2000; Ernestus & Baayen 2003). We found both of these deletion processes productive, counter to the predictions of Wheeler (2005) and Pons-Moll (2015), and further that this productivity involves frequency-matching: the relative deletion rates observed in production (Experiment 1) and the ratings in the acceptability judgement task (Experiment 2) matched the lexically-observed order *frequent suffix* > *other* > *monosyllabic stems* > *penultimately-stressed stems*. However, at the level of detail there were discrepancies: notably, speakers were less likely to delete /r/ compared to /n/. We conjectured that this is due either to exposure to dialects without /r/-deletion, or to the fact that /n/-deletion is spelt.

Our second research question concerned the productivity of “unnatural” phonology in the form of opacity and saltation. For opacity, we examined the interaction of /n/-deletion with /nt/-cluster simplification, with which it is in a counterfeeding relationship. Counter to the predictions of Sanders (2003), the opaque pattern /n/ → ∅, /nt/ → [n] does seem to be productive: it was repeatedly volunteered in Experiment 1 and found acceptable in Experiment 2. Simple “opacity repair” with feeding (both /nt/ and /n/ surface as ∅) was extremely rare; rather the main alternative response was outright suppression of /nt/-cluster simplification. The latter is somewhat of a puzzle since this process is exceptionless in the lexicon. We argued that the lowered rates of /nt/-cluster simplification may be related to the very fact that it is part of an opaque interaction, but other possible explanations include exposure to /nt/-retaining dialects or to orthography.

Saltation, seen in the alternation of [ʒ] with [tʃ], showed somewhat similar results. Participants did provide saltatory responses about half of the time in Experiment 1, and rated them highly in Experiment 2. However, [ʃ]-final forms, with “saltation repair,” were also frequently offered in Experiment 1, despite the fact that forms with repaired saltation are not attested in the lexicon nor present in other dialects. This is consistent with the hypothesis that saltation is “unnatural phonology,” liable to repair (White 2014; Hayes & White 2015), though low attestation competes as a possible explanation.

Beyond the purely phonological questions we had in mind in designing the experiment, the results proved informative in a different way: our participants were often “avoidant,” in the sense that they used unusual morphology to dodge the question at hand regarding phonology. We suggest that avoidance is related to uncertainty, as shown by the correlation between avoidant response with weak lexical support and also with long response times.

5.2 Directions for future work

The outcomes of our experiment included cases in which our participants gave responses that mismatch the pattern of their lexicon. For several of these, we speculatively offered multiple explanations, but nothing in our data permits us to distinguish between these explanations. For instance, it seems impossible at present to know whether the large number of saltation repairs we observed resulted from a bias against saltation itself, from their sparse lexical attestation, or indeed from some factor as yet unnoticed.

One route toward resolving these issues would be to appeal to typology. Ideally, we would want to look at many further cases in which the combination of available explanations is different, in order to sort out the explanations that are truly effective. To give just one example from existing work, in Seediq phonology, the saltation $e \rightarrow u$ (saltating over i ; Kuo 2020, 2023) is abundantly attested in the lexicon. Kuo tested the productivity of this pattern experimentally and found that participants never “repaired” saltation, i.e., producing $e \rightarrow i$. At present, very few such examples can be cited, because only a few phonological systems have been wug-tested.

We also feel that more formal work on orthographic influence in phonology is called for. Above, we conjectured that Catalan orthography, which spells out /n/-deletion and /ʒ/ ~ [tʃ] saltation but not /r/-deletion and /nt/-cluster simplification, may have been a factor in our participants’ responses. Specifically, participants deleted /r/ less than /n/, perhaps because the deleted /r/, unlike /n/, is reflected by a silent letter (e.g., [ˈklar-ə] ~ [ˈkla] is spelt *clara* ~ *clar*; [ˈsan-ə] ~ [ˈsa] is spelt *sana* ~ *sa*); similarly, the willingness of participants to accept [nt] as the output for /nt/, despite its zero lexical attestation, may have been influenced by the fact that the sequence is rendered orthographically by *nt* (e.g., [ˈsant-ə] ~ [ˈsan] is spelt *santa* ~ *sant*). We see it as a possible topic for future research to develop a formal model that accounts for such influences. It

should be quite feasible to construct an “orthographic phonology” for Catalan, which would recapitulate, using letters, the processes of /n/-deletion and /z/ ~ [tʃ] saltation—but not the processes of /r/-deletion and /nt/-cluster simplification. This orthographic phonology could be augmented with models that account for how Catalan speakers contrive orthographic forms for novel words they hear, and contrive phonological forms to match a given orthographic form. These models must all be coupled, perhaps probabilistically, to account for how both phonology and orthographic systems contribute to participant’s responses.

Lastly, we would look forward to formal work addressing the origin of speaker uncertainty. We suggested (§4.3) that uncertainty is the most likely cause of avoidant responses; empirically, we found that such responses are most common for phenomena that are poorly attested in the lexicon. That poor attestation should create uncertainty seems completely intuitive to us; however, it is not a prediction made by current constraint-based learning models (e.g., MaxEnt, or other frameworks like Stochastic OT, Boersma 1998). Such models frequency-match even when this matching is based on very few forms. One might imagine future models that incorporate uncertainty by allocating probability mass to a “Don’t Know” candidate, but research is needed to find a principled way to do this.

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